



Committee Document

**IEC/ISO JSEG 15/WS
1 N1**

Metaverse — Vocabulary

Metaverse — Vocabulaire

Edition 0.4



COPYRIGHT PROTECTED DOCUMENT

© ISO/IEC 2024

All rights reserved. Unless otherwise specified, or required in the context of its implementation, no part of this publication may be reproduced or utilized otherwise in any form or by any means, electronic or mechanical, including photocopying, or posting on the internet or an intranet, without prior written permission. Permission can be requested from either ISO at the address below or ISO's member body in the country of the requester.

ISO copyright office
CP 401 • Ch. de Blandonnet 8
CH-1214 Vernier, Geneva
Phone: +41 22 749 01 11
Email: copyright@iso.org
Website: www.iso.org

Published in Switzerland

Contents

Page

Foreword.....iv

Introduction.....v

1 Scope.....1

2 Normative references.....1

3 Terms and definitions.....1

3.1 Metaverse core concepts.....1

3.2 Metaverse technology domains.....4

3.3 XR concepts.....4

3.4 Metaverse interfaces.....5

3.7 Metaverse spatial.....6

3.8 Extended metaverse concepts.....7

3.9 Governance.....7

4 Definition of Metaverse.....7

4.1 General.....7

4.2 Approach.....7

4.3 Background.....7

4.4 Considerations.....8

4.5 Consensus.....8

Annex A Abbreviated terms.....9

Bibliography.....10

Index.....12

Foreword

ISO (the International Organization for Standardization) and IEC (the International Electrotechnical Commission) form the specialized system for worldwide standardization. National bodies that are members of ISO or IEC participate in the development of International Standards through technical committees established by the respective organization to deal with particular fields of technical activity. ISO and IEC technical committees collaborate in fields of mutual interest. Other international organizations, governmental and non-governmental, in liaison with ISO and IEC, also take part in the work.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of document should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives)

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO and IEC shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents) or the IEC list of patent declarations received (see <https://patents.iec.ch>).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see www.iso.org/iso/foreword.html. In the IEC, see www.iec.ch/understanding-standards.

This document was prepared by ISO/IEC JSEG 15, *Metaverse*.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html and www.iec.ch/national-committees.

Introduction

This document presents terms and definitions for concepts relevant to the Metaverse.

Specifically, the terminology presented in this document:

- serve as a sound basis in the understanding of Metaverse;
- guide new developments in the field by underpinning mutual understanding; and
- serve as a quick and handy reference for those newly introduced to this field.

Metaverse — Vocabulary

1 Scope

This document provides a concept system and a standard set of terminology related to the Metaverse, from basic concepts to those of the Metaverse ecosystem.

The scope of the vocabulary provided in this document corresponds to that of ISO/IEC JSEG 15.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <https://www.electropedia.org>

3.1 Metaverse core concepts

3.1.1

metaverse

interactive (3.1.15) virtual world (3.1.21) that incorporates spatial referencing (3.1.20) experienced through immersive (3.1.14) technologies

3.1.2

artificial intelligence

AI

capacity of computers or other machines to exhibit or simulate intelligent behavior

[SOURCE: IEEE Std 7010™-2020^[24]]

3.1.3

assistive technology

piece of equipment, product system, hardware, software or service that is used to enable, maintain or improve functional capabilities of individuals with disabilities

[SOURCE: ITU-T F.791^[26]]

3.1.4

cloud computing

paradigm for enabling network access to a scalable and elastic pool of shareable physical or virtual resources with self-service provisioning and administration on-demand

[SOURCE: ISO/IEC 17788:2014, 3.2.5]

3.1.5

decentralized identifier

DID

identifier in the format of the W3C Decentralized Identifier

[SOURCE: W3C did-core^[28]]

3.1.6

digital human

realistic virtual human representations created using advanced simulation technology

3.1.7

digital twin

digital representation of an object of interest

Note 1 to entry: A digital twin may require different capabilities (e.g., synchronization, real-time support) according to the specific domain of application.

[SOURCE: ITU-T Y.4600^[27]]

3.1.8

distributed computing

model of computing in which a set of nodes coordinates its activities by means of digital messages passed between the nodes

[SOURCE: ISO/IEC TR 23188:2020, 3.1.1]

3.1.9

distributed ledger

a *ledger* ([3.1.18](#)) that is shared across a set of DLT nodes and synchronized between the DLT nodes using a consensus mechanism

[SOURCE: ISO 22739:2024, 3.22]

3.1.10

digital asset

asset that exists only in digital form or which is the digital representation of another asset

[SOURCE: ISO 24165-1:2021, 3.9]

3.1.11

digital identity

information used by information systems to represent an external agent, a person, organization, application, or device

3.1.12

edge computing

distributed computing ([3.1.8](#)) in which processing and storage takes place at or near the edge, where the nearness is defined by the system's requirements

[SOURCE: ISO/IEC TR 23188:2020, 3.1.3]

3.1.13

edge device

device that provides an entry point into enterprise or service provider core networks

EXAMPLE Gateways, routers, switches, multiplexers, and a variety of other access devices.

[SOURCE: W3C wot-architecture[30]]

3.1.14

immersive

creating the illusion of being inside a computer-generated scene

[SOURCE: ISO/IEC 19775-1:2023]

3.1.15

interactive

pertaining to a system or mode of operation in which each user entry causes a response from or action by the system

[SOURCE: ISO/IEC/IEEE 24765:2017, 3.2050, modified — removed second definition.]

3.1.16

Internet of Things

system of entities (including cyber-physical devices, information resources, and people) that exchange information and interact with the physical world by sensing, processing information, and actuating

[SOURCE: IEEE Std 2413™-2019^[23], modified — dropped article.]

3.1.17

interoperability

ability of two or more systems or applications to exchange information and to mutually use the information that has been exchanged

[SOURCE: ISO/IEC 17788:2014, 3.1.5]

3.1.18

ledger

information store that keeps records of transactions that are intended to be final, definitive and immutable

[SOURCE: ISO 22739:2024, 3.43]

3.1.19

information security

preservation of confidentiality, integrity and availability of information

Note 1 to entry: In addition, other properties, such as authenticity, accountability, non-repudiation, and reliability can also be involved.

[SOURCE: ISO/IEC 27000:2018, 3.28]

3.1.20

spatial referencing

method to describe a position within a context

Note 1 to entry: A position may the form of a label, code or coordinate tuple.

Note 2 to entry: Definition modified from ISO 19111:2019, 3.1.56 “spatial reference”.

3.1.21

virtual world

cyber world

simulated environment that can be accessed by multiple users through an online interface

Note 1 to entry: The simulated environments are often interactive.

Note 2 to entry: The physical world in which people live, and the related characteristics, will be referred to as the “real world” to differentiate it from a virtual world.

[SOURCE: ISO/IEC 27032:2012, 4.51]

3.1.22

virtual office

3-D virtual scene replicating office space of an organization, providing users with an *immersive* ([3.1.14](#)) experience

3.1.23

user-generated content

UGC

content created by users within the *metaverse* (3.1.1)

Note 1 to entry: UGC is often facilitated by simplified editing tools.

3.2 Metaverse technology domains

3.2.1

industrial metaverse

metaverse (3.1.1) that integrates virtual and physical realities to create a sophisticated digital industrial ecosystem, enhancing real-world processes through virtual simulations and strengthening the entire industrial and value chain with increased intelligence, collaboration, openness, service orientation, and interconnectivity

3.2.2

financial metaverse

metaverse (3.1.1) enabling a new type of financial business where the physical world and the mirrored digital world are symbiotic

3.2.3

education metaverse

metaverse (3.1.1) enabling a multi-sensory *immersive* (3.1.14) classroom connecting virtual and reality, enabling knowledge acquisition, participation, and creation

3.2.4

healthcare metaverse

metaverse (3.1.1) that enables the digital, intelligent and accurate medical information platform that breaks space-time restrictions and technical barriers of the original medical system

3.2.5

eCommerce metaverse

metaverse (3.1.1) that enables marketing towards users, creating personalized, emotional and social marketing scenes through metaverse technology and content

3.2.6

game metaverse

metaverse (3.1.1) that enables the evolution of gaming that embodies four characteristics: immersion, accessibility, accessibility and extensibility

3.2.7

social metaverse

metaverse (3.1.1) that creates a holographic virtual reality that integrates the advantages of online and offline social networking, using *AR* (3.3.2), *VR* (3.3.4), *XR* (3.3.1) and other technologies

3.3 XR concepts

3.3.1

extended reality

XR

environments experienced through human-machine sensory interactions commonly composed of real world and virtual elements

Note 1 to entry: Extended reality covers *virtual reality* (3.3.4), *augmented reality* (3.3.2) and *mixed reality* (3.3.3) environments.

3.3.2

augmented reality

AR

interactive experience of a real-world environment whereby the objects that reside in the real world are augmented by computer-generated perceptual information

[SOURCE: ISO/IEC 18038:2020, 3.1]

3.3.3

mixed reality

MR

display continuum in which both real and virtual images are combined in some way and in some proportion

Note 1 to entry: Augmented reality and virtual reality are considered to be on the mixed reality continuum.

[SOURCE: ISO/IEC TR 23843:2020, 3.4, modified — added Note 1 to entry.]

3.3.4

virtual reality

VR

artificial environment presented in the computer

[SOURCE: ISO/IEC TR 18121:2015, 3.6, modified — dropped article]

3.3.5

mixed reality system

system that uses a mixture of representations of physical world data and *virtual world* ([3.1.21](#)) data as its presentation medium

[SOURCE: ISO/IEC 18039:2019, 3.1.13]

3.3.6

mixed and augmented reality

MAR

integration of real and virtual worlds including *mixed reality* ([3.3.3](#)) and *augmented reality* ([3.3.2](#))

Note 1 to entry: MAR is often used interchangeably with MR.

[SOURCE: ISO/IEC 18038:2020]

3.4 Metaverse interfaces

3.4.1

cyber physical system

CPS

system with digital, analogue, cyber, physical and human components interacting with each other, engineered to function through integrated physics and logic

[SOURCE: IWA 39:2022, 3.1^[25]]

3.5

human-computer interface

HCI

boundary across which physical interaction between a human being and a computing device takes place

Note 1 to entry: Definition originates from ISO/IEC TR 10000-3:1998, 3.2.5.

3.6

brain-computer interface

BCI

device that quantifies central nervous system activity and translates them into digital outputs

Note 1 to entry: A brain-computer interface may facilitate bi-directional communication between the brain and a computing device.

Note 2 to entry: A brain-computer interface is a type of human-computer interface that enables direct communication between the brain and an external device.

3.6.1

haptics

input or output device that senses or actuates the body's movements by means of physical contact with the user

[SOURCE: ISO/IEC 23093-1:2022, 3.1.5]

3.6.2

game console

dedicated electronic device designed to play video games

3.6.3

internet of things

IoT

infrastructure of interconnected entities, people, systems and information resources, together with services which process and react to information from the physical world and the virtual world

[SOURCE: ISO/IEC 20924:2021, 3.2.4]

3.6.4

wearable device

IoT ([3.6.3](#)) device designed for operation near to, on, or inside of a body

Note 1 to entry: Wearable devices often have a variety of sensing abilities, but limited power capacity constraining communication and data processing abilities. As critical devices of the IoT, it is considered that the communication between wearable devices and a network might not require any human intervention. Wearable devices include electronic devices usable by humans, animals, and other organisms.

[SOURCE: ISO/IEC 20924:2024, 3.1.35]

3.7 Metaverse spatial

3.7.1

simultaneous localization and mapping

SLAM

technique used in robotics and computer vision to create a map of an unknown environment while simultaneously keeping track of an agent's location within it

3.7.2

neural radiance fields

NeRF

method for synthesizing novel views of complex scenes by optimizing a continuous 3D scene representation

3.7.3

photogrammetry

technique of measurement using photographs, for example aerial photographs, to determine, primarily, geometric properties such as size, location and form of objects

Note 1 to entry: Photogrammetric measurement is often used for mapping, but also has some engineering applications.

[SOURCE: ISO 7078:2020, 3.1.5]

3.7.4

computer vision

artificial vision

capability of a functional unit to acquire, process, and interpret visual data

Note 1 to entry: Computer vision involves the use of visual sensors to create an electronic or digital image of a visual scene.

[SOURCE: ISO/IEC 2382:2015, Clause 2123787, modified — Note to entries 2-4 omitted.]

3.7.5

computer graphics

methods for converting data to or from graphic displays by a computer

[SOURCE: ISO 6707-2:2017, 3.2.17]

3.7.6

point cloud

discrete set of data points in space

Note 1 to entry: A point cloud typically representing a 3D shape or object.

3.8 Extended metaverse concepts

3.8.1

C-Meta

low-carbon social metaverse under the Web 3.0 architecture, aiming to build a new green and energy-saving digital economy

3.9 Governance

3.9.1

stakeholder

role, position, individual, organization, or classes thereof, having an interest, right, share, or claim, in an entity of interest

EXAMPLE End users, operators, acquirers, owners, suppliers, architects, developers, builders, maintainers, regulators, taxpayers, certifying agencies, and markets.

[SOURCE: ISO/IEC/IEEE 42010:2022, 3.17, modified — dropped article]

3.9.2

stakeholder perspective

way of thinking about an entity of interest, especially as it relates to concerns

[SOURCE: ISO/IEC/IEEE 42010:2022, 3.18, modified — The original example has been removed.]

4 Definition of Metaverse

4.1 General

The definition of a “metaverse” has long been subject of controversy.

4.2 Approach

In order to define any concept, it is necessary to determine the essential characteristics of the concept, where additions or subtractions to these characteristics will no longer define the concept.

4.3 Background

The word “metaverse” originated from the science fiction *Snow Crash* written by Neal Stephenson in 1992 Reference [34].

In popular culture, the term “metaverse” is commonly used as a generalized entity beyond the Internet, as a massively multi-player virtual world experienced through immersive technologies with comprehensive sensory integration through human computer interfaces.

The group has considered a few popular definitions of the “metaverse”, including:

- From Reference [31]: “universal, immersive virtual world that is facilitated by virtual reality (VR) and augmented reality (AR)”
- From Reference [32]: persistent and immersive simulated world that is experienced in the first person by large groups of simultaneous users who share a strong sense of mutual presence
- From Reference [33]: an infinite and persistent set of computer-generated virtual worlds that may be accessed via some human computer interface

4.4 Considerations

WS 1 has considered various definitions, common perspectives and expectations on the term “metaverse” and has come to the following conclusions based on consensus.

Essential characteristics of the “metaverse” include:

virtual	as in the items in the experience is virtual
immersive	where it is experienced through human-computer interaction
spatial	where the experience is spatially organized according to some notion of location
interactive	where users can directly interact with it and receive feedback

Common characteristics expected by popular culture include:

massively multi-user	where there can be many simultaneous users
interconnection	of virtual world systems
interoperability	between users across virtual world systems
persistent	where actions and events in the virtual world persists after a user leaves
real-time	where user actions are performed in real-time

4.5 Consensus

The resulting consensus arrived at this definition for “metaverse”:

metaverse: interactive virtual world that incorporates spatial referencing experienced through immersive technologies

Annex A **(informative)**

Abbreviated terms

AGI	artificial general intelligence
AI	artificial intelligence
API	application programming interface
AR	augmented reality
DLT	distributed ledger technology
IoT	Internet of Things
ML	machine learning
OMF	Open Modeling Foundation
OWL	Web Ontology Language
RDF	Resource Description Framework
UDT	Urban Digital Twin
VR	virtual reality
WoT	Web of Things
W3C	World Wide Web Consortium
XR	collective reference to both AR and VR

Bibliography

- [1] ISO 6707-2:2017, *Buildings and civil engineering works — Vocabulary — Part 2: Contract and communication terms*
- [2] ISO 7078:2020, *Buildings and civil engineering works — Procedures for setting out, measurement and surveying — Vocabulary*
- [3] ISO 19111:2019, *Geographic information — Referencing by coordinates*
- [4] ISO 22739:2024, *Blockchain and distributed ledger technologies — Vocabulary*
- [5]¹⁾ ISO 24165-1:2021, *Digital token identifier (DTI) — Registration, assignment and structure — Part 1: Method for registration and assignment*
- [6] ISO/IEC 2382:2015, *Information technology — Vocabulary*
- [7] ISO/IEC TR 10000-3:1998, *Information technology — Framework and taxonomy of International Standardized Profiles — Part 3: Principles and Taxonomy for Open System Environment Profiles*
- [8]²⁾ ISO/IEC 17788:2014, *Information technology — Cloud computing — Overview and vocabulary*
- [9] ISO/IEC 18038:2020, *Information technology — Computer graphics, image processing and environmental representation — Sensor representation in mixed and augmented reality*
- [10] ISO/IEC 18039:2019, *Information technology — Computer graphics, image processing and environmental data representation — Mixed and augmented reality (MAR) reference model*
- [11] ISO/IEC TR 18121:2015, *Information technology — Learning, education and training — Virtual experiment framework*
- [12] ISO/IEC 19775-1:2023, *Computer graphics, image processing and environmental data representation — Extensible 3D (X3D) — Part 1: Architecture and base components*
- [13]³⁾ ISO/IEC 20924:2021, *Information technology — Internet of Things (IoT) — Vocabulary*
- [14] ISO/IEC 20924:2024, *Internet of Things (IoT) and digital twin — Vocabulary*
- [15]⁴⁾ ISO/IEC 23093-1:2022, *Information technology — Internet of media things — Part 1: Architecture*
- [16] ISO/IEC TR 23188:2020, *Information technology — Cloud computing — Edge computing landscape*
- [17] ISO/IEC TR 23843:2020, *Information technology for learning, education and training — Catalogue model for virtual, augmented and mixed reality content*
- [18] ISO/IEC/IEEE 24765:2017, *Systems and software engineering — Vocabulary*
- [19] ISO/IEC 27000:2018, *Information technology — Security techniques — Information security management systems — Overview and vocabulary*

1) Withdrawn.

2) Withdrawn.

3) Withdrawn.

4) Withdrawn.

- [20]⁵⁾ ISO/IEC 27032:2012, *Information technology — Security techniques — Guidelines for cybersecurity*
- [21] ISO/IEC/IEEE 42010:2022, *Software, systems and enterprise — Architecture description*
- [22] ITU FGMV-O-235, ITU-T. *Draft Technical Specification ITU FGMV — Vocabulary for metaverse*
- [23]⁶⁾ IEEE Std 2413™-2019, INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS. *IEEE Standard for an Architectural Framework for the Internet of Things (IoT)*. Updated May 1, 2020. Approved. Institute of Electrical and Electronics Engineers
- [24]⁷⁾ IEEE Std 7010™-2020, INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS. *IEEE Recommended Practice for Assessing the Impact of Autonomous and Intelligent Systems on Human Well-Being*. Approved. Institute of Electrical and Electronics Engineers
- [25] IWA 39:2022, INTERNATIONAL ORGANIZATION FOR STANDARDIZATION. *Gap analysis for standardization on sustainable and human-centred societies enabled with cyber physical systems*. Version 1. 60
- [26]⁸⁾ ITU-T F.791, INTERNATIONAL TELECOMMUNICATION UNION. *Accessibility terms and definitions*. Version 2. Published. International Telecommunication Union
- [27]⁹⁾ ITU-T Y.4600, INTERNATIONAL TELECOMMUNICATION UNION. *Requirements and capabilities of a digital twin system for smart cities*. Version 1. Published. International Telecommunication Union
- [28]¹⁰⁾ W3C did-core, *Decentralized Identifiers (DIDs) v1.0*. World Wide Web Consortium
- [29] W3C did-spec-registries, [NO INFORMATION AVAILABLE]
- [30] W3C wot-architecture, [NO INFORMATION AVAILABLE]
- [31] ISO/IEC JTC 1. *JTC 1 Standards and Standardization for the Metaverse* [website]. 2023. Available at: <https://jtc1info.org/jtc-1-standards-and-standardization-for-the-metaverse/>
- [32] ROSENBERG, L. *Metaverse 101: defining the key components* [website]. VentureBeat. 2022. Available at: <https://venturebeat.com/business/metaverse-101-defining-the-key-components/>
- [33] STOTHARD, P., P. RYAN and T. KURATA. *Mining Metaverse Taxonomy (NOT YET PUBLISHED)*. 2024
- [34] NEAL STEPHENSON. *Snow Crash*. Bantam Books

5) Withdrawn.

6) Available at <https://ieeexplore.ieee.org/document/9032420>

7) Available at <https://ieeexplore.ieee.org/document/9084219>

8) Available at <https://handle.itu.int/11.1002/1000/13661>

9) Available at <https://handle.itu.int/11.1002/1000/15073>

10) Available at <https://www.w3.org/TR/did-core/>

Index

AGI, _____
AI, [1](#), _____
API, _____
AR, [5](#), _____
artificial intelligence, [1](#)
assistive technology, [1](#)
augmented reality, [4](#)
BCI, [5](#)
brain-computer interface, [5](#)
C-Meta, [7](#)
cloud computing, [1](#)
computer graphics, [7](#)
computer vision, [6](#)
CPS, [5](#)
cyber physical system, [5](#)
decentralized identifier, [1](#)
DID, [2](#)
digital asset, [2](#)
digital human, [2](#)
digital identity, [2](#)
digital twin, [2](#)
distributed computing, [2](#)
distributed ledger, [2](#)
DLT, _____
eCommerce metaverse, [4](#)
edge computing, [2](#)
edge device, [2](#)
education metaverse, [4](#)
extended reality, [4](#)
financial metaverse, [4](#)
game console, [6](#)
game metaverse, [4](#)
haptics, [6](#)
HCI, [5](#)
healthcare metaverse, [4](#)
human-computer interface, [5](#)
immersive, [2](#)
industrial metaverse, [4](#)
information security, [3](#)
interactive, [3](#)
internet of things, [6](#)
interoperability, [3](#)
IoT, _____
ledger, [3](#)
MAR, [5](#)
metaverse, [1](#)
mixed and augmented reality, [5](#)
mixed reality, [5](#)
mixed reality system, [5](#)
ML, _____
MR, [5](#)
NeRF, [6](#)
neural radiance fields, [6](#)
OMF, _____
OWL, _____
photogrammetry, [6](#)
point cloud, [7](#)
RDF, _____
simultaneous localization and mapping, [6](#)
SLAM, [6](#)
social metaverse, [4](#)
spatial referencing, [3](#)
stakeholder, [7](#)
stakeholder perspective, [7](#)
UDT, _____
user-generated content, [4](#)
virtual office, [3](#)
virtual reality, [5](#)
virtual world, [3](#)
VR, [5](#), _____
W3C, _____
wearable device, [6](#)
WoT, _____
XR, [4](#), _____



Price based on 11 pages

© IEC/ISO 2024
All rights reserved

iso.org